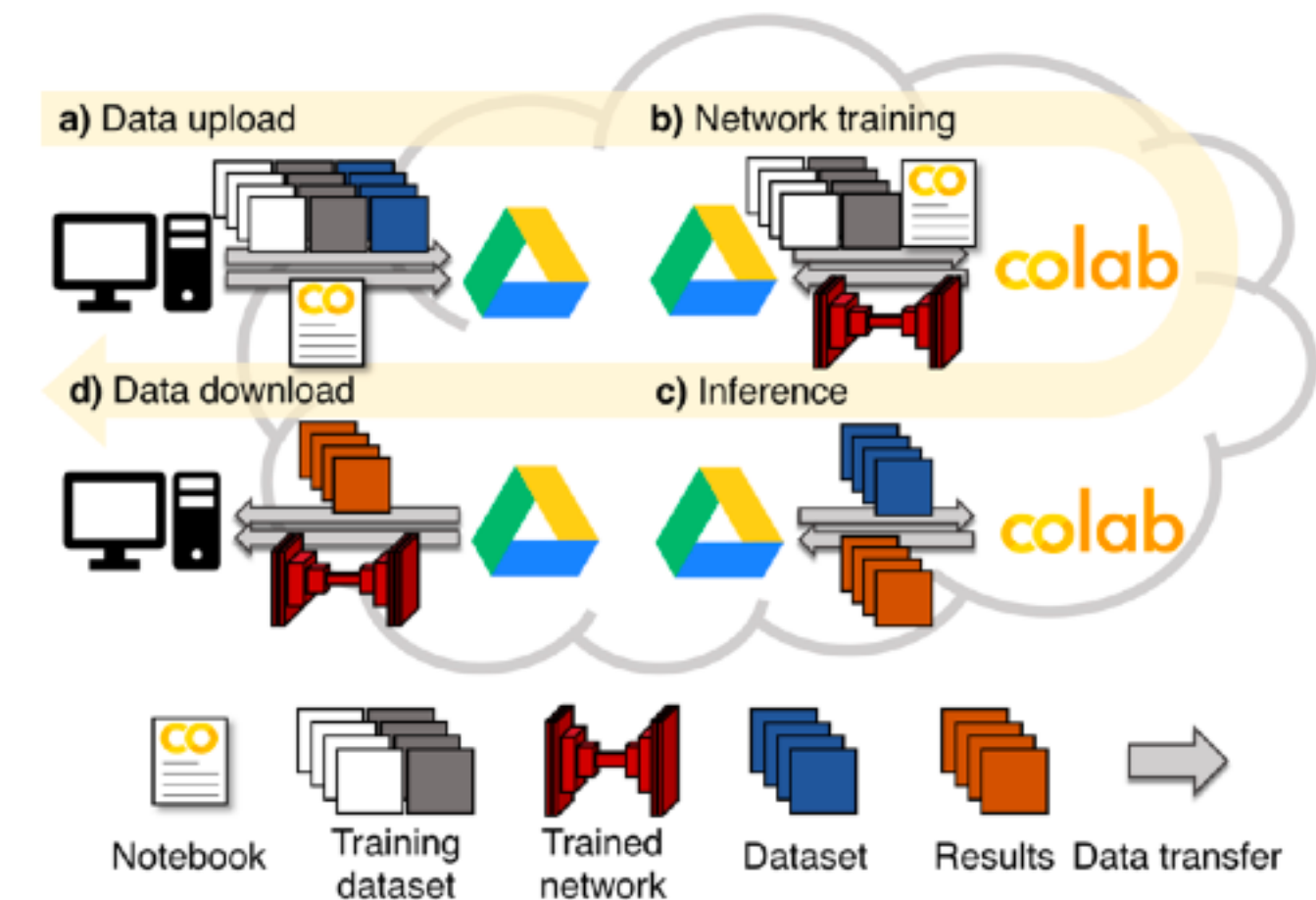


Bioimage analysis 4 years ago (2020)

- CellPose is out (2020)
 - “Cellpose: a generalist algorithm for cellular segmentation”
 - Trained on highly varied images of cells, over 70,000 segmented objects
 - Cells don't have to be star-shaped
 - Web-platform and Jupyter notebooks
 - <https://www.nature.com/articles/s41592-020-01018-x>
- ZeroCostDL4Mic available
 - [Paper](#) still a year in the future
 - Implementation of common DL technologies to microscopy imaging
 - Relies on GPUs and other infrastructure provided by Google Colab.
 - <https://github.com/HenriquesLab/ZeroCostDL4Mic/wiki>



Bioimage analysis now (2024)

- Increased integration of DL technologies into existing bioimage tools
 - CARE, Stardist, CellPose, [DeepCell](#) (MESMER) as plugins, modules, notebooks for
 - Fiji, CellProfiler, [Napari](#), ZeroCostDL4Mic, etc
- Most major Microscopy producers offer AI
 - Implemented in-house or through acquisition
 - Zeiss: Arivis (bought)
 - Leica: Aivia (bought)
 - Nikon: NIS.ai suite (in-house implementation of open-source technologies)
- “Smart microscopy” includes DL for event detection
 - <https://www.nature.com/articles/s41592-022-01589-x>
- Fully autonomous cars still in the future
 - Image analysis is hard if you don't understand what you are looking at